RO2 - Lecture 4
3D-Accelerometer (Part 2)

we focus on students

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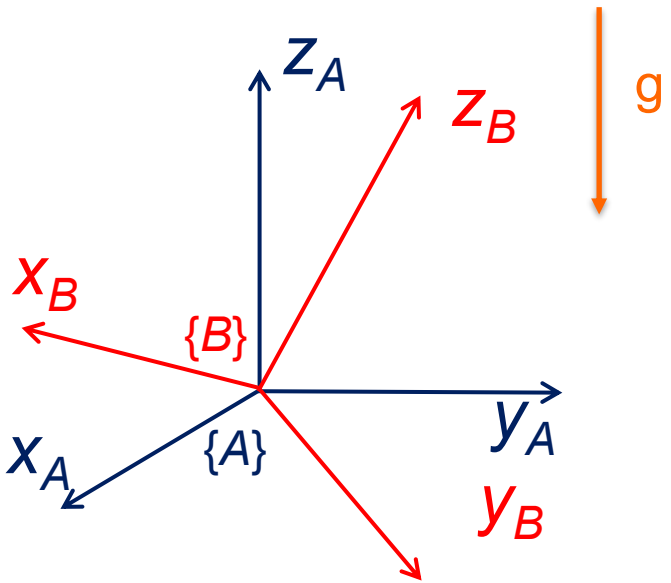
Determining the roll and pitch angle based on acceleration measurements



known: $\boldsymbol{a}_m = {}^{B}\boldsymbol{g}$, ${}^{A}\boldsymbol{g} = \begin{bmatrix}0\\0\\1\end{bmatrix}^T$

$${}^{B}\boldsymbol{g} = {}^{B}\boldsymbol{a}_{\mathrm{m}} = {}^{A}\boldsymbol{R}_B^{-1}\,{}^{A}\boldsymbol{a}_{\mathrm{m}} = {}^{A}\boldsymbol{R}_B^{-1}\,{}^{A}\boldsymbol{g} = {}^{A}\boldsymbol{R}_B^{-1}\begin{bmatrix}0\\0\\1\end{bmatrix}$$

Repetition

Goal:
Determine roll and pitch based on accelerometer reading

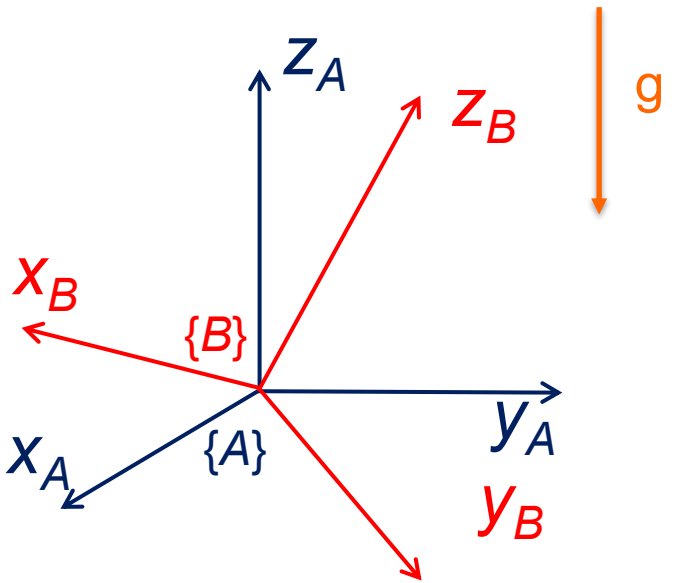
Approach:
1. Find mathematical expression for ${}^{A}\boldsymbol{R}_B^{-1}$ as combinations of R_x, R_y and R_z
2. Solve for roll and pitch angles

Result:

$$\varphi = \operatorname{atan2}\left(\boldsymbol{a}_{norm_y}, \boldsymbol{a}_{norm_z}\right)$$

$$\theta = \operatorname{atan2}\left(-\boldsymbol{a}_{norm_x}, \sqrt{\boldsymbol{a}_{norm_y}^2 + \boldsymbol{a}_{norm_z}^2}\right)$$

Describing the quaternion representing rotation from frame A to B



known: $\boxed{\boldsymbol{a}_m = {}^{B}\boldsymbol{g}}$, ${}^{A}\boldsymbol{g} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}^T$

$${}^{A}\boldsymbol{q}_{\boldsymbol{g}} = \begin{bmatrix} 0 & {}^{A}\boldsymbol{g} \end{bmatrix}$$

$${}^{B}\boldsymbol{q}_{\boldsymbol{g}} = \begin{bmatrix} 0 & \boxed{{}^{B}\boldsymbol{g}} \end{bmatrix}$$

$${}^{B}\boldsymbol{q}_{\boldsymbol{g}} = {}^{B}\boldsymbol{q}_A \, {}^{A}\boldsymbol{q}_{\boldsymbol{g}} \, {}^{B}\boldsymbol{q}_A^* = \begin{bmatrix} 0 \\ \boxed{\begin{matrix} 2(q_1 q_3 + q_0 q_2) \\ 2(q_2 q_3 - q_0 q_1) \\ q_0^2 - q_1^2 - q_2^2 + q_3^2 \end{matrix}} \end{bmatrix}^T$$

Goal:
Find a quaternion ${}^{B}\boldsymbol{q}_A$ that rotates ${}^{A}\boldsymbol{q}_{\boldsymbol{g}} = \begin{bmatrix} 0 & {}^{A}\boldsymbol{g} \end{bmatrix}$ into ${}^{B}\boldsymbol{q}_{\boldsymbol{g}}$ by fulfilling the equation above

Using:

$${}^{B}\boldsymbol{q}_A = \begin{bmatrix} q_0 & q_1 & q_2 & q_3 \end{bmatrix} \quad \text{and} \quad {}^{A}\boldsymbol{q}_{\boldsymbol{g}} = \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^{B}\boldsymbol{q}_A^* = \begin{bmatrix} q_0 & -q_1 & -q_2 & -q_3 \end{bmatrix}$$

Describing the quaternion representing rotation from frame A to B

- Goal: given measurement vector $\boldsymbol{a}_m$, find ${}^{B}\boldsymbol{q}_{A}$ based on:

$$\boldsymbol{a}_m = \begin{bmatrix} a_{m_x} \\ a_{m_y} \\ a_{m_z} \end{bmatrix}^T = \begin{bmatrix} 2(q_1 q_3 + q_0 q_2) \\ 2(q_2 q_3 - q_0 q_1) \\ q_0^2 - q_1^2 - q_2^2 + q_3^2 \end{bmatrix}^T$$

- 3 known parameters $a_{m_x}, a_{m_y}, a_{m_z}$, 4 unknown parameters q_0, q_1, q_2, q_3 → underdetermined
- Infinite solutions, as rotation of gravity vector in the measurement contains no information regarding *z*-axis

Describing the quaternion representing rotation from frame A to B

- Goal: given measurement vector $\boldsymbol{a}_m$, find ${}^{B}\boldsymbol{q}_A = [q_0\ q_1\ q_2\ q_3]$ which fulfills:

$$\boldsymbol{a}_m = \begin{bmatrix} a_{m_x} \\ a_{m_y} \\ a_{m_z} \end{bmatrix}^T = \begin{bmatrix} 2(q_1 q_3 + q_0 q_2) \\ 2(q_2 q_3 - q_0 q_1) \\ q_0^2 - q_1^2 - q_2^2 + q_3^2 \end{bmatrix}^T$$

- Applying $-\boldsymbol{a}_m$ on both sides yields:

$$\underbrace{\begin{bmatrix} 2(q_1 q_3 + q_0 q_2) \\ 2(q_2 q_3 - q_0 q_1) \\ q_0^2 - q_1^2 - q_2^2 + q_3^2 \end{bmatrix}^T - \begin{bmatrix} a_{m_x} \\ a_{m_y} \\ a_{m_z} \end{bmatrix}^T}_{\text{Objective function } f(\boldsymbol{a}_m,\ {}^{B}\hat{\boldsymbol{q}}_A)} = 0$$

- One possibility: iteratively find solution for ${}^{B}\hat{\boldsymbol{q}}_A$ which minimizes $f(\boldsymbol{a}_m,\ {}^{B}\hat{\boldsymbol{q}}_A)$, i.e. find solution to:

$$\min_{{}^{B}\hat{\boldsymbol{q}}_A} f\left(\boldsymbol{a}_m,\ {}^{B}\hat{\boldsymbol{q}}_A\right)$$

Using a *gradient descent algorithm* for finding a quaternion solution

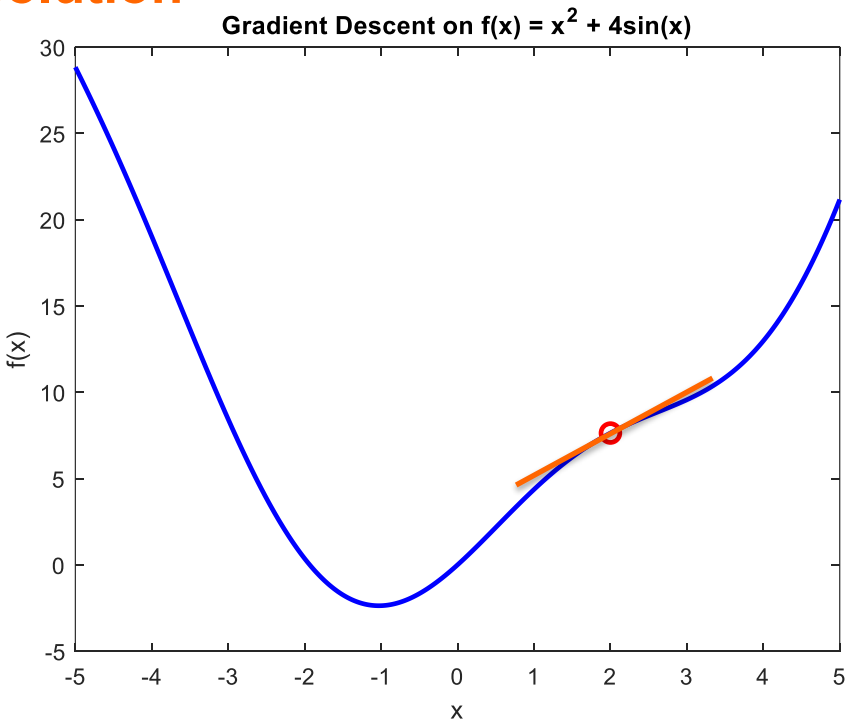
- The gradient of a function points in the direction of the *steepest* increase
- General approach of gradient descent algorithm:

  Find the solution by taking repeated steps in the *opposite* direction of the gradient of the objective function at the current point.

- Given a multi-variable objective function $F(\boldsymbol{x})$ and an initial solution guess $\boldsymbol{x}_0$, iterative solutions are found by:

$$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k - \mu \nabla F(\boldsymbol{x}_k)$$

where $\nabla F(\boldsymbol{x}_k)$ is the gradient of the objective function at point $\boldsymbol{x}_k$, and μ is the step size



Simple example for gradient descent algorithm

- Example: searching the minimum of the following function with single variable x :

$$F(x) = x^2 + 4\sin(x)$$

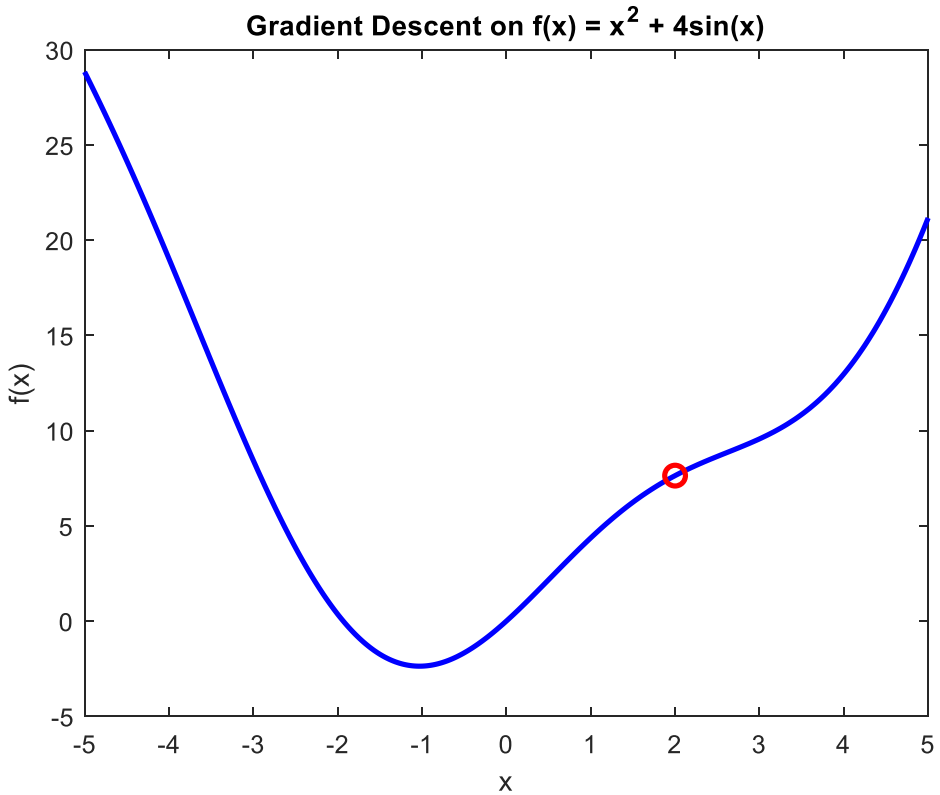
- Gradient $\nabla F(x)$:

$$\nabla F(x) = F'(x) = 2x + 4\cos(x)$$

- Step size μ = 0.1, threshold $\mathrm{ths} = 10^{-2}$
- Stop criterion: $|x_k - x_{k-1}| < \mathrm{ths}$

Simple example for gradient descent algorithm

$f(x) = x^2 + 4 \cdot \sin(x)$



objective function:

$F(x) = f(x) = x^2 + 4 \cdot \sin(x)$

Goal: $\min_{\hat{x}} F(\hat{x})$

$$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k - \mu \nabla F(\boldsymbol{x}_k)$$

Gradient: $\nabla F(x) = 2x + 4\cos(x)$

Step size: μ = 0.1

Initial step:

$x = x_k = x_0 = 2$

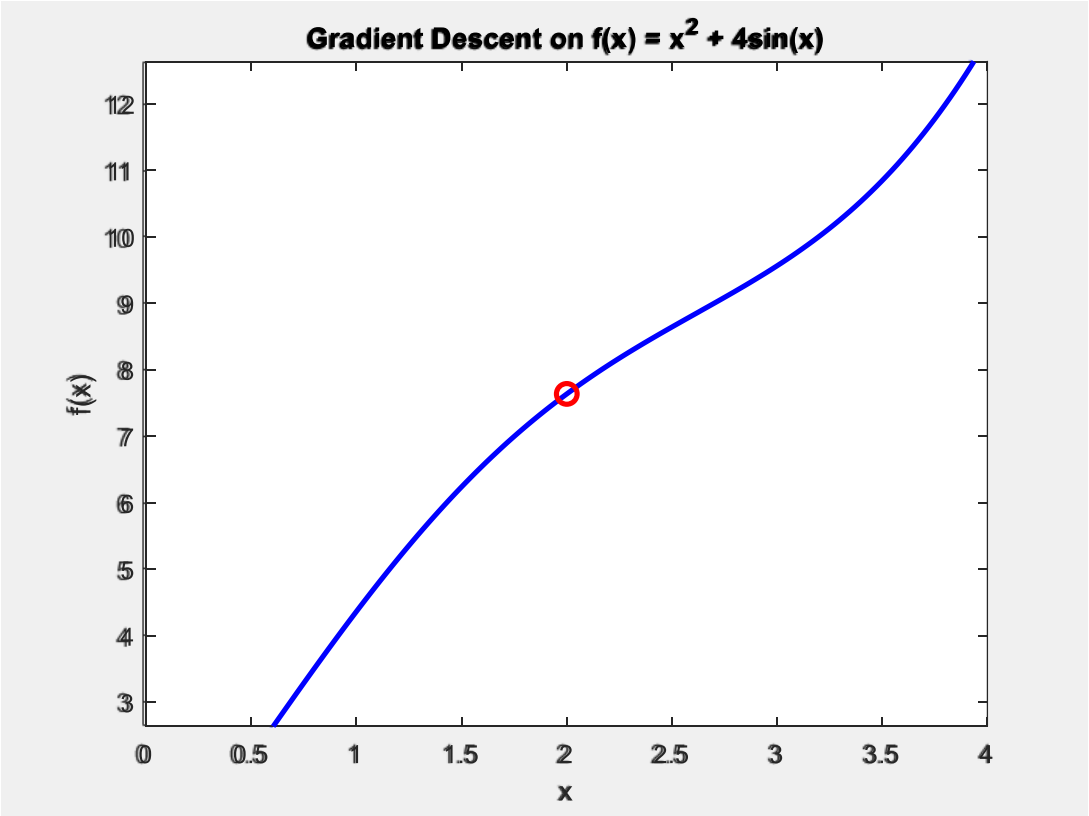
$\nabla F(x_0) = 2x_0 + 4\cos(x_0) = 2.3354$

$x_{k+1} = x_k - \mu \nabla F(x_k)$

$x_{0+1} = x_0 - 0.1 \cdot \nabla F(x_0)$

$x_1 = 1.7665$

Simple example for gradient descent algorithm



$$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k - \mu \nabla F(\boldsymbol{x}_k)$$

$$\nabla F(x) = 2x + 4\cos(x)$$

$\mu = 0.1$

Using a *gradient descent algorithm* for finding a quaternion solution

- Stop criteria:
  - maximum number of iterations exceeded, i.e. $k \geq k_{max}$
  - norm of the gradient falls below a threshold, i.e.

$$\| \nabla F(x_k) \| < \varepsilon$$

  - change of value of objective function falls bellow a threshold, i.e.

$$|F(\boldsymbol{x}_k) - F(\boldsymbol{x}_{k-1})| < \text{threshold}$$

  - change of iteration value (i.e.: the optimization variable(s)) falls bellow a threshold, i.e.

$$|x_k - x_{k-1}| < \text{threshold}$$

Using a *gradient descent algorithm* for finding a quaternion solution

- Non-linear relation between acceleration and quaternion:

$$G\left({}^{B}\boldsymbol{q}_{A}, \boldsymbol{a}_m\right) = {}^{B}\boldsymbol{q}_{A} - \boldsymbol{a}_m = \begin{bmatrix} 2(q_1q_3 + q_0q_2) \\ 2(q_2q_3 - q_0q_1) \\ q_0^2 - q_1^2 - q_2^2 + q_3^2 \end{bmatrix}^T - \begin{bmatrix} a_{m_x} \\ a_{m_y} \\ a_{m_z} \end{bmatrix}^T$$

- For readability ${}^{B}\boldsymbol{q}_{A} = \boldsymbol{q}$ and as $\boldsymbol{a}_m$ is constant for every optimization cycle:

$$G(\boldsymbol{q}) = \begin{bmatrix} 2(q_1q_3 + q_0q_2) - a_{m_x} \\ 2(q_2q_3 - q_0q_1) - a_{m_y} \\ q_0^2 - q_1^2 - q_2^2 + q_3^2 - a_{m_z} \end{bmatrix}^T = \begin{bmatrix} G_x(\boldsymbol{q}) \\ G_y(\boldsymbol{q}) \\ G_z(\boldsymbol{q}) \end{bmatrix}^T$$

- $G(\boldsymbol{q})$ is a residual **vector**, but scalar required → squared error norm is chosen as objective function:

$$F(\boldsymbol{q}) = \frac{1}{2} G(\boldsymbol{q})^T G(\boldsymbol{q})$$

Using a *gradient descent algorithm* for finding a quaternion solution - summary

$$\boldsymbol{q}_{k+1} = \boldsymbol{q}_k - \mu \nabla F(\boldsymbol{q}_k)$$

$$\nabla F(\boldsymbol{q}) = \boldsymbol{J}_G(\boldsymbol{q})^T G(\boldsymbol{q})$$

$$\boldsymbol{J}_G(\boldsymbol{q}) = \begin{matrix} \frac{\delta G_x(\boldsymbol{q})}{\delta q_0} & \frac{\delta G_x(\boldsymbol{q})}{\delta q_1} & \frac{\delta G_x(\boldsymbol{q})}{\delta q_2} & \frac{\delta G_x(\boldsymbol{q})}{\delta q_3} \\ \frac{\delta G_y(\boldsymbol{q})}{\delta q_0} & \frac{\delta G_y(\boldsymbol{q})}{\delta q_1} & \frac{\delta G_y(\boldsymbol{q})}{\delta q_2} & \frac{\delta G_y(\boldsymbol{q})}{\delta q_3} \\ \frac{\delta G_z(\boldsymbol{q})}{\delta q_0} & \frac{\delta G_z(\boldsymbol{q})}{\delta q_1} & \frac{\delta G_z(\boldsymbol{q})}{\delta q_2} & \frac{\delta G_z(\boldsymbol{q})}{\delta q_3} \end{matrix}$$

3x1

$$G(\boldsymbol{q}) = \begin{bmatrix} 2(q_1 q_3 + q_0 q_2) - a_{m_x} \\ 2(q_2 q_3 - q_0 q_1) - a_{m_y} \\ q_0^2 - q_1^2 - q_2^2 + q_3^2 - a_{m_z} \end{bmatrix}$$

3x4

$$\boldsymbol{J}_G = \begin{bmatrix} 2q_2 & 2q_3 & 2q_0 & 2q_1 \\ -2q_1 & -2q_0 & 2q_3 & 2q_2 \\ 2q_0 & -2q_1 & -2q_2 & 2q_3 \end{bmatrix}$$

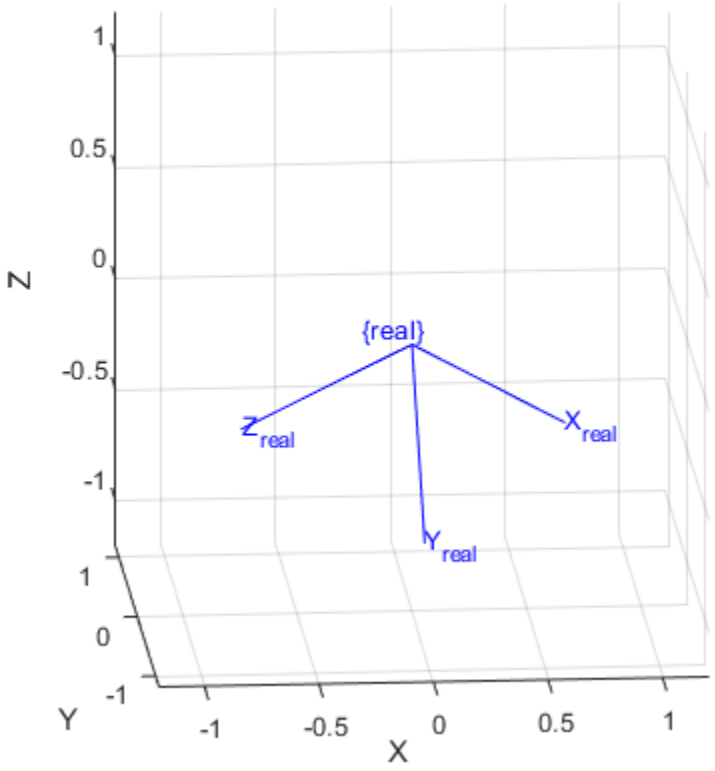
ex4_3_gradient_descent.m

Exercise L4.3 – Quaternion from accelerometer reading

- Download and the file ex4_3_quaternion_gradient_descent.m and check the implementation
- Play around with the start values used for the gradient descent algortihm → what do you observe?

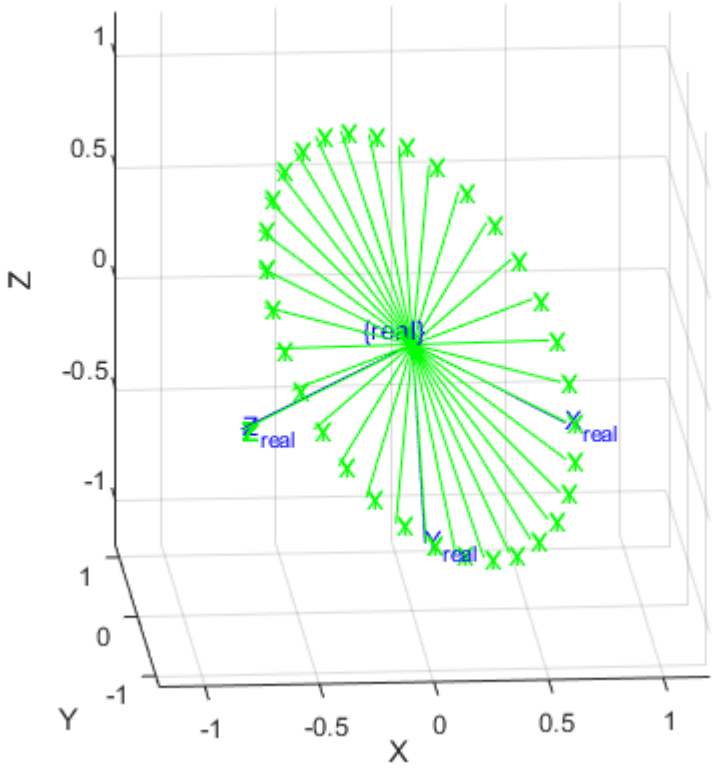
Ambiguity regarding *z*-axis rotation

- E.g.: this figure shows the axes for a given 3D-orientation



Ambiguity regarding *z*-axis rotation

- E.g.: this figure shows the axes for a given 3D-orientation
- Solutions found for the estimated orientation based on different start values are shown in green



Summary

+ Orientation estimation does not require initial knowledge

+ Quality of orientation estimation is good under stationary conditions

+ Roll/Pitch angles, or quaternion with arbitrary *z*-axis orientation can be determined

- 3D accelerometers can be used to determine rotation about the global *x*- and *y*-axis, but not about the *z*-axis (rotation parallel to the axis parallel to gravity vector)

- Bad quality of orientation estimation under dynamic conditions